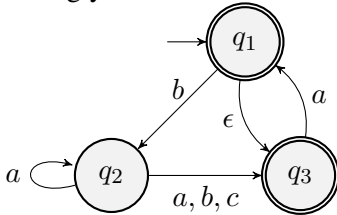


CS410 Final Part (a)

5 January 2023

I. MULTIPLE CHOICE QUESTIONS (50 POINTS)

Given the following NFA M_1 . Answer question 1 accordingly.



1. How many rows and column does the start transition table have?

- a) Rows: 3, Columns: 3
- b) Rows: 3, Columns: 4
- c) Rows: 4, Columns: 3
- d) Rows: 4, Columns: 4
- e) None

2. How many states do we need at minimum if we want to design a DFA for the following language $L = \{w|w \text{ contains exactly three a's}\}$?

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5

3. Which of the following language is defined with the regular expression $\Sigma\Sigma$?

- a) $A = \{w|w \text{ is a string of odd length}\}$
- b) $B = \{w|w \text{ is a string of even length}\}$
- c) $C = \{w| \text{length of } w \text{ is a multiple of } 2\}$
- d) $D = \{w| \text{length of } w \text{ is } 2\}$
- e) None

4. Which of the following is not a **standard** regular expression operator?

- a) *
- b) $\cap$
- c) $\cup$
- d) $\circ$
- e) All of the above are **standard** regular operators

5. Given the language $A = \{0^n 1^n | n \geq 0\}$, and p the pumping length. Let say we wanted to prove that A is not regular. Let say the selected string is $0^p 1^p$. Which of the following could be the selected string y , that can be pumped?

- a) 1
- b) 0^{p+1}
- c) 1111
- d) 00
- e) 01

6. Which of the following languages is not regular?

- a) $A = \{a^{2n} b^m c | n, m \geq 0\}$
- b) $B = \{a^{3n+2} b^{m+1} | m, n \geq 0\}$
- c) $C = \{a^n b^{3n} | n \text{ is odd}\}$
- d) $D = \{a^k b^m c^n | k \geq 2, m \geq 0, n \leq 2\}$
- e) All of them are regular

7. Which of the following languages is regular?

- a) $A = \{0^n 1^n | n \geq 0\}$
- b) $B = \{w|w \text{ shows a binary number divisible by } 5\}$
- c) $C = \{w|w \text{ has an equal number of 0s and 1s}\}$
- d) $D = \{ww|w \in \{0, 1\}^*\}$
- e) None

8. Which of the following is the regular expression for the language $A = \{w| \text{every 0 in } w \text{ is followed by at least one 1}\}$?

- a) $\Sigma 01 \Sigma$
- b) $\Sigma 01^+ \Sigma$
- c) $(01^+)^*$
- d) $\Sigma^* 01^+ \Sigma^*$
- e) $\Sigma^* (01^+)^* \Sigma^*$

9. Which of the following CFG generates the language $L = \{a^n b^m | n > m\}$?

- a) $S \rightarrow aSb \mid T$
 $T \rightarrow aa$
- b) $S \rightarrow aSb \mid a$
- c) $S \rightarrow aSb \mid T$
 $T \rightarrow aT \mid a$
- d) $S \rightarrow aSb \mid T$
 $T \rightarrow aTb \mid a \mid b$
- e) All of them correct

10. Which of the following languages is ambiguous, that is it has at least two different leftmost derivations?

a)

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T * F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

b)

$$E \rightarrow E + E \mid E * E \mid a$$

c)

$$\begin{aligned} S &\rightarrow 0S \mid 1A \\ A &\rightarrow 0A \mid \epsilon \end{aligned}$$

d) All of them are ambiguous

e) None

11. Given the following ambiguous CFG, which of the following strings has two different leftmost derivations?

$$\begin{aligned} S &\rightarrow AB \mid BA \\ A &\rightarrow A0 \mid \epsilon \\ B &\rightarrow 1B \mid \epsilon \end{aligned}$$

a) 111

b) 11

c) 00

d) All of them

e) None

12. If we have 11 variables on the right hand side of a rule R , how many rules should we add as extra instead of R if we want to convert R into CNF?

a) 9

b) 10

c) 11

d) 12

e) 13

13. Which of the following languages is context-free and regular?

a) $L = \{a^n b^n \mid n \geq 0\}$

b) $L = \{a^m b^n \mid n \geq 0, m \geq 0\}$

c) $L = \{a^n b^{2n} \mid n \geq 0\}$

d) $L = \{a^{2n} b^n \mid n \geq 0\}$

e) None of them

14. Which of the following CFG is in CNF?

a)

$$\begin{aligned} S &\rightarrow 0 \mid AB \mid CD \\ C &\rightarrow AB \\ D &\rightarrow A \\ A &\rightarrow 0 \\ B &\rightarrow 1 \end{aligned}$$

b)

$$\begin{aligned} S &\rightarrow AC \mid CC \mid C \\ E &\rightarrow DC \\ C &\rightarrow BB \\ D &\rightarrow BD \mid DA \mid a \mid b \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

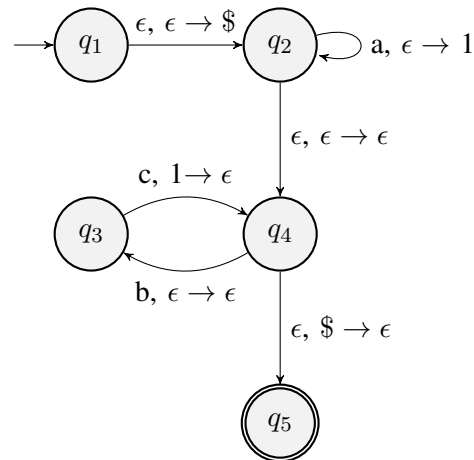
c)

$$\begin{aligned} S &\rightarrow AA \mid A \mid 0 \mid 1 \mid \epsilon \\ A &\rightarrow 0 \\ B &\rightarrow 1 \end{aligned}$$

d) All of them in CNF

e) None of them in CNF

Given the following PDA P_1 . Answer question 14 accordingly.



15. How many columns does the start transition table have?

a) 16

b) 15

c) 12

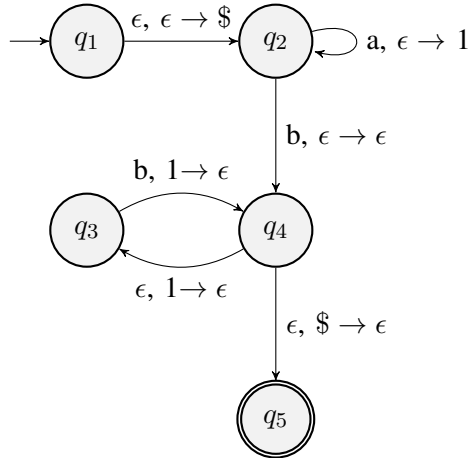
d) 9

e) 8

16. Which of the following languages is not context-free?

- a) $L = \{a^{2n}b^n \mid n \geq 1\}$
- b) $L = \{a^n b^m c^m d^n \mid n \geq 0\}$
- c) $L = \{a^n b^{2m} c^{2n} d^{2m} \mid n, m \geq 0\}$
- d) $L = \{a^n b^n \mid n \geq 0\}$
- e) All of them not context-free

17. What is the language generated by the following PDA?



- a) $L = \{a^n b^n \mid n \geq 0\}$
- b) $L = \{a^n b^{n+1} \mid n \geq 0\}$
- c) $L = \{a^n b^{2n+1} \mid n \geq 0\}$
- d) $L = \{a^{2n} b^{n+1} \mid n \geq 0\}$
- e) $L = \{a^n b^{2n} \mid n \geq 0\}$

18. Which of the following string is a possible configuration for a Turing machine?

- a) 101101101
- b) $q_0 1011 q_2 01101$
- c) 101101101, q_2
- d) $1011 q_2 01101$
- e) None of the above

19. If we apply the state transition function $\delta(q_i, b) = (q_j, c, L)$ to the configuration $ubq_i b v$, which configuration will we have next?

- a) $uq_j b c v$
- b) $uq_j c b v$
- c) $uq_i c b v$
- d) $u b c q_j v$
- e) None of the above

20. Which of the following languages is not decidable?

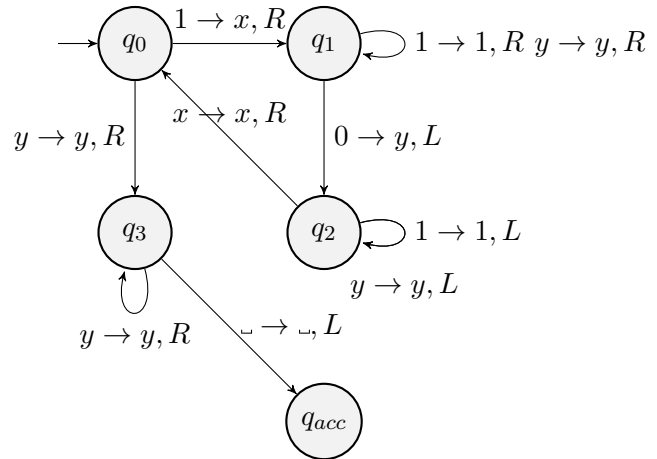
- a) $A = \{ \langle A, B \rangle \mid A \text{ and } B \text{ are CFG, } L(A) = L(B) \}$
- b) $B = \{ \langle G, w \rangle \mid G \text{ is a CFG that generates string } w \}$
- c) $C = \{ \langle G \rangle \mid G \text{ is a DFA and } L(G) = \emptyset \}$
- d) Every context free language
- e) All of the above

21. Which of the following sets is the smallest?

- a) Regular languages
- b) Context-free languages
- c) Decidable languages
- d) Turing-recognizable languages
- e) All of them equal in size

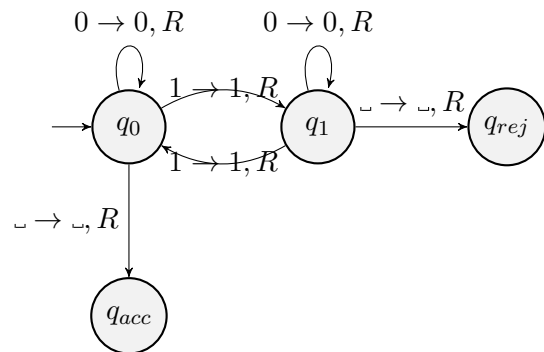
22. Which of the following languages is not countable?

- a) $\mathbb{Z}$
- b) Even numbers
- c) $\mathbb{N}$
- d) $\mathbb{Q}$
- e) All of them are countable



23. Which of the following languages is depicted in the TM given above?

- a) $L = \{0^{2n}1^n \mid n \geq 1\}$
- b) $L = \{0^n1^n \mid n \geq 1\}$
- c) $L = \{0^n1^n0^n \mid n \geq 1\}$
- d) $L = \{1^n0^n \mid n \geq 1\}$
- e) $L = \{1^{2n}0^n \mid n \geq 1\}$



24. Which of the following languages is depicted in the TM given above?

- a) $L = \{w \mid \text{Number of 1's in } w \text{ is a multiple of 3}\}$
- b) $L = \{0^n1^n \mid n \geq 1\}$
- c) Language of palindromes of even length
- d) $L = \{1^n0^n \mid n \geq 1\}$
- e) $L = \{w \mid \text{Number of 1's in } w \text{ is a multiple of 2}\}$